Project Title	**Pneumonia Detection from Chest X-Ray Images using CNN**
Skills take away From This Project	**Deep Learning Fundamentals and Deployment of the Model**
Domain	**Deep Learning**

Problem Statement:

This project challenges learners to build an image classification model capable of detecting pneumonia from chest X-ray images. The model should accurately differentiate between healthy lungs and those affected by pneumonia.

Business Use Cases:

- **Clinical Decision Support:** Assisting doctors in early and accurate pneumonia diagnosis, especially in resource-limited settings.
- **Telemedicine:** Enabling remote diagnosis in rural or underserved areas.
- **Hospital Triage Automation:** Automatically flagging suspected pneumonia cases from X-rays for further examination.
- **Medical Imaging Software:** Integrating AI-based diagnostic tools into hospital PACS (Picture Archiving and Communication Systems).

Approach:

Learners will follow a machine learning workflow as outlined below:

1. **Load and explore the dataset.**

2. **Preprocess the images** (e.g., resizing, normalization, train-validation-test split).

3. **Build and train a CNN** from scratch or using transfer learning (e.g., MobileNet, VGG).

4. **Evaluate the model** using accuracy, confusion matrix, and classification metrics.

5. **Visualize results** and understand model performance.

6. **Optimize the model** for improved accuracy and robustness.

Results:

- A trained CNN model capable of classifying X-ray images as *Pneumonia* or *Normal*.

- Visualizations of training performance, confusion matrix, and sample predictions.

Project Evaluation metrics:

- **Accuracy** – Overall correct predictions.

- **Precision, Recall, F1 Score** – Especially important due to class imbalance.

- **Confusion Matrix** – For detailed error analysis.

- **Model Robustness** – Ability to generalize on unseen data.

Technical Tags:

`Deep Learning`, `CNN`, `Image Classification`, `Medical Imaging`, `Pneumonia Detection`, `TensorFlow`, `Keras`, `Healthcare AI`, `Computer Vision`

Data Set:

- **Name:** Chest X-Ray Images (Pneumonia)
- **Source:** Kaggle: Chest X-Ray Images (Pneumonia)
- **Format:** JPEG image files organized into `train`, `val`, and `test` folders
- **Classes:** `PNEUMONIA`, `NORMAL`

Data Set Explanation:

- The dataset contains chest X-ray images labeled as either *PNEUMONIA* or *NORMAL*.
- Images are stored in separate folders for training, validation, and testing.

Preprocessing Steps:

- Resize all images to a consistent size (e.g., 150x150 or 224x224).
- Normalize pixel values (scale to 0-1).
- Apply data augmentation (optional) to improve generalization.

Project Deliverables:

- **Source Code:** Python scripts or Jupyter Notebooks for model training and evaluation.
- **Trained Model:** A ready to deploy model
- **Visuals:** Training plots, confusion matrix, sample prediction outputs.

Project Guidelines:

- Include clear comments and modular code structure.
- Validate results using multiple metrics.
- Handle class imbalance with techniques like weighted loss or oversampling if needed.
- Ensure reproducibility with random seeds and environment specification.

Timeline:

Week	Milestone
Day 1	Dataset exploration and preprocessing completed
Day 2 & 3	Model built and initial training done
Day 4	Model tuning, evaluation, and visualizations
Day 5	Final deliverables prepared and submitted

Project Title	**Next Word Prediction Using Recurrent Neural Networks (Language Modeling)**
Skills take away From This Project	**Deep Learning Fundamentals and Deployment of the Model**
Domain	**Deep Learning**

Problem Statement: The goal of this project is to build a language model that can predict the next word in a sequence using RNN-based architectures like LSTM or GRU. This task will help learners understand how sequential data is handled in NLP.

Business Use Cases:

- Autocompletion in Text Editors: Suggest the next word as the user types.
- Voice Assistants: Improve speech-to-text accuracy by predicting contextually appropriate words.
- Search Engines: Predict search queries for faster input.
- Machine Translation & Chatbots: Serve as a building block for encoder-decoder models.

Approach:

1. Load and explore the dataset (e.g., WikiText-2).
2. Preprocess text data: tokenization, vocabulary creation, padding.
3. Create sequences of input words and corresponding target words.
4. Build a language model using an RNN or LSTM
5. Train the model to predict the next word in the sequence.

6. Evaluate model performance.
7. Generate sample text using the trained model.

Results:

- A trained RNN-based model capable of predicting the next word given a context.
- Quantitative metrics to evaluate model fluency.

Project Evaluation Metrics:

- Accuracy – Proportion of correctly predicted next words.
- Loss curves – To assess learning over epochs.

Technical Tags:

NLP, Language Modeling, RNN, LSTM, GRU, Next Word Prediction, Text Generation, Deep Learning, TensorFlow, PyTorch

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Dataset:

- Name: WikiText-2
- Source: Hugging Face Datasets
- Format: Plain text files
- Variables: Articles split into tokens, structured for word-level prediction

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Dataset Explanation:

- WikiText-2 is a cleaned subset of Wikipedia articles.

- Rich in language structure, ideal for training word-level language models.
- Preprocessing Steps:
    - Tokenization
    - Lowercasing
    - Padding/truncation to uniform sequence length
    - Splitting into training and validation sets

Project Deliverables:

1. Source Code: Python scripts or Jupyter notebooks with preprocessing, model training, and evaluation.
2. Model File: A ready to deploy model

Project Guidelines:

- Use appropriate preprocessing for sequence data.
- Document hyperparameters and training observations.
- Try both basic RNNs and advanced variants like LSTM

Timeline:

Week	Milestone
Day 1	Dataset loading and preprocessing completed
Day 2,3	Sequence generation and RNN model implementation
Day 4	Training, evaluation, and text generation
Day 5	Final documentation and submission of deliverables

Project Title	**Exploratory Data Analysis on Global Superstore Sales Dataset**
Skills take away From This Project	**Data Preprocessing Fundamentals**
Domain	**Data Preprocessing**

Problem Statement:

This project aims to develop learners' skills in data understanding and insight generation by performing comprehensive EDA on a retail sales dataset. The goal is to identify patterns, trends, and anomalies in sales, profits, and customer behavior across regions and categories.

Business Use Cases:

- **Retail Strategy Planning:** Identify profitable regions, product categories, and customer segments.
- **Inventory Optimization:** Analyze sales patterns to optimize stock levels.
- **Revenue Forecasting:** Discover trends in order frequency, profit margins, and delivery times.
- **Customer Behavior Analysis:** Understand customer demographics and buying habits for targeted marketing.

Approach:

1. **Load the dataset** and understand its structure (variables, types, missing values).

2. **Clean the data**: handle missing values, duplicates, and outliers.
3. **Univariate Analysis:** Distribution of numerical and categorical variables.
4. **Bivariate and Multivariate Analysis:** Correlations, group-wise comparisons.
5. **Time Series Patterns:** Analyze order trends over months and years.
6. **Visualizations:** Use plots (histograms, boxplots, heatmaps, bar charts, line plots) to convey insights.
7. **Key Findings Summary:** Interpret results with business implications.

Results:

- Identification of best- and worst-performing products, categories, and regions.
- Insights into profit margins, shipping methods, and discount impact.
- Clear, interactive visualizations to support business decisions.

Project Evaluation Metrics:

- **Depth and Breadth of Analysis** – Coverage of key variables and relationships.
- **Clarity of Visualizations** – Use of appropriate charts and labels.
- **Insightfulness** – Relevance and utility of the insights drawn.
- **Code Quality** – Clean, organized, well-commented code.
- **Presentation** – Structured storytelling in notebooks or reports.

Technical Tags:

EDA, Data Visualization, Data Cleaning, Pandas, Matplotlib, Seaborn, Plotly, Retail Analytics, Business Intelligence, Python

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Dataset:

- **Name:** Global Superstore Dataset
- **Source:** [Kaggle – Global Superstore]
- **Format:** CSV file
- **Variables:** Order Date, Ship Mode, Segment, Region, Category, Sales, Quantity, Discount, Profit

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Dataset Explanation:

- The dataset contains sales and order data for a global retail company.
- Each row represents a single order line item with attributes like customer info, product details, sales, and profit.
- **Preprocessing Steps:**
  - Parse date fields
  - Handle missing and inconsistent values
  - Remove duplicates and irrelevant features (if any)

Project Deliverables:

1. **Jupyter Notebook or Python Script**: With step-by-step EDA.
2. **Data Cleaning Script**: Reusable preprocessing functions.
3. **Visualizations**: Plots and dashboards in notebooks.

Project Guidelines:

- Maintain a logical flow: data load → clean → explore → visualize → summarize.
- Use libraries like pandas, matplotlib, seaborn, plotly.
- Ensure all charts have titles, axis labels, and legends where necessary.
- Comment code for clarity and reproducibility.

Timeline:

Week	Milestone
Day 1	Data loading and cleaning completed
Day 2	Univariate and bivariate analysis with visualizations
Day 3	Multivariate and temporal analysis; insight generation
Day 4	Final report, code polishing, and project submission